Dcard: Providing engaging social networking services with Google Cloud

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About Dcard

Founded in 2011, Dcard is a leading online social networking platform provider in Taiwan. Starting as a networking service dedicated to college students, Dcard has grown to become one of the most popular social media platforms among Taiwanese young people while expanding its business into ecommerce, Intellectual Property content and advertising. Currently, Dcard has 20 million monthly unique visitors and 8 million members across Taiwan, Hong Kong, Macau, and Japan.

Google Cloud results

- Shortens software development cycles with microservices in GKE
- Helps save data storage costs through diverse storage classes of Cloud Storage
- Supports near real-time data analytics for quick service improvements and troubleshooting with BigQuery
- Enables the development of a content recommendation system serving 20 million monthly unique visitors within three months

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Industries: Media & Entertainment

Location: Taiwan

Functions
(https://cloud.google.com/functions/), Cloud

CloudMile

About CloudMile

As one of the leading AI and cloud service providers in Taiwan, CloudMile empowers businesses to accelerate digital transformation through cloud technology and machine learning. As of November 2023, the company holds more than 200 professional certifications and has served 700+ enterprise customers. CloudMile is a Google Cloud partner specializing in infrastructure, machine learning, and data analytics.

- Google Cloud (https://cloud.google.com/)
- BigQuery (https://cloud.google.com/bigquery/)
- Cloud Data Loss Prevention (https://cloud.google.com/dlp/)
- Cloud Functions (https://cloud.google.com/functions/)
- Cloud Run (https://cloud.google.com/run/)
- Cloud SQL (https://cloud.google.com/sql/)

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platform Dcard (https://www.dcar

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Compute Engine (https://cloud.google.com/compute/)

percent (https://www.businessweekly.com.t ag_page.aspx?id=7006493)

infrastructure, machine learning, and data

Cloud Dataproc (https://cloud.google.com/dataproc/)

among people aged between 18

country, the platform is one of th

Kubernetes Engine (https://cloud.google.com/kubernetes-engine/)

the younger generation in Taiwan

topics and seek information or a

Cloud Pub/Sub (https://cloud.google.com/pubsub/)

on a range of topics surrounding everyday life from outfits to relationships.

Dcard's founding team first launched the platform in 2011 as a networking service dedicated to college students. As the popularity of its forum grew, the startup expanded its business to include ecommerce, Intellectual Property content, and advertising services on its platform. Over the past few years, Dcard has also entered markets abroad for users in Hong Kong, Macau and Japan. As of November 2023, Dcard had 20 million monthly unique visitors and 8 million members in total.

"Our mission is to provide an online space where everyone can find something they're interested in. That's why we've been constantly adding engaging features to our platform," explains Taco, Engineering Manager at Dcard. "For example, many users on Dcard have expressed a desire for more diverse ways to convey their emotions. As a response, we have launched the original IP 'dtto friends' stickers. This enables everyone to respond to posts with relatable stickers, adding a new dimension to self-expression."

To ensure its platform continues to provide reliable services with its growing popularity, Dcard started to look for an IT infrastructure with higher scalability in 2016 to deploy its social networking platform in Taiwan. The team chose Google Cloud (https://cloud.google.com/) because it was the only public cloud provider that had a data center in Taiwan at the time, enabling dozens of milliseconds less latency for users in the country compared to other offerings.

"Our online forum was growing very fast in Taiwan, so we needed a highly scalable IT infrastructure to provide stable services," says Taco. "Google Cloud not only supports high scalability but also lower latency, which can help us ensure our service quality."

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A highly scalable cloud infrastructure with flexible resources

The Dcard team moved its online forum to Google Cloud by leveraging microservices in Google Kubernetes Engine (https://cloud.google.com/gke) (GKE) and virtual machines (VMs) in Compute Engine (https://cloud.google.com/compute) as servers, and Cloud Storage (https://cloud.google.com/storage) to back up data and store video content. With the timely technical support and comprehensive advice on product selection from the Google Cloud partner CloudMile (https://mile.cloud/), Dcard was able to quickly resolve issues and continue improving its operations when expanding its business.

"With its profound knowledge in cloud technology, the CloudMile team understands our pain points very well and can therefore effectively help us overcome technical challenges. Although our team is already familiar with the Google Cloud tools, we still encounter setting errors from time to time, and that's when CloudMile's expertise and timely responses come in handy," says Taco.

One major advantage of using Google Cloud for the Dcard team is the diverse computing resources available. For its application services, Dcard relies on the microservice architecture supported by GKE to shorten development cycles and the autoscaling feature of GKE to ensure stable services despite exponential traffic growth when popular events like FIFA World Cup take place. For feature testing, it uses Cloud Functions (https://cloud.google.com/functions) and Cloud Run (https://cloud.google.com/run) to simplify deployment and reduce costs. On top of that, being able to easily create isolated projects in Google Cloud has also helped the team at Dcard develop multiple new features at the same time without affecting existing services.

The high flexibility of resource use also applies to data storage. Since Cloud Storage provides various storage classes, Dcard can choose the most suitable storage plans based on the nature of its data and lower the overall costs. For example, it uses the archive storage class to store historical log data, which would only be used for auditing, and the coldline storage class to store backup data, which are occasionally retrieved for data analyses.

"The cloud infrastructure of Google Cloud not only delivers great performance, but also provides us with flexible resources to support our operations in the most cost-effective way possible," notes Taco. "We're therefore able to offer highly reliable services and develop more new features."

Enabling quick service improvements and troubleshooting with BigQuery

To provide engaging social networking services and support data-driven decision-making, Dcard needs to analyze terabytes of data every day. It employs Pub/Sub (https://cloud.google.com/pubsub) to transfer data from its platforms to its data warehouse in BigQuery (https://cloud.google.com/bigquery) and other databases for analytics. Taco notes that since Pub/Sub possesses great buffer capacity, the team is able to stream or batch insert all the data it needs. The fast query speed supported by BigQuery also enables Dcard to realize near real-time analytics. This means that it can identify and resolve issues more timely or even in advance.

"We're dedicated to providing engaging social networking services, so we need to know our users' preferences very well, which is why data analytics is crucial to our business," he adds. "With the powerful analytics capabilities of BigQuery, we're able to quickly find out what our users like or dislike through A/B testing, and effectively improve our services and operations based on near real-time insights."

Setting up its data warehouse in Google Cloud has also helped Dcard implement solid data governance. The team leverages the single sign-on authentication scheme in Google Cloud to control data access for better protection of sensitive data, and Cloud Data Loss Prevention (https://cloud.google.com/dlp/) to automatically identify data with high security risks.

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Building a content recommendation system to boost user engagement

Using its data pipeline established in Google Cloud as foundation, the Dcard team has successfully developed a content recommendation system API within three months. It uses Dataproc (https://cloud.google.com/dataproc/) and GKE to build an ETL tool that can transform raw data in BigQuery into features for machine learning (ML) model training. These features are then stored in Cloud SQL (https://cloud.google.com/sql/), with its High Availability configuration activated to protect important data. To enhance cost-effectiveness, Dcard trains its ML models using custom computing resources in Compute Engine, and deploys the recommendation API in GKE, which helps realize high scalability with minimal manual work required.

"With the diverse and abundant computing and data tools of Google Cloud, we managed to build a content recommendation system serving 20 million monthly unique visitors in a short period of time," says Taco. "Now, our users can see recommended content that is highly aligned with their interests on the main page of our online forum, which has boosted user engagement."

Supporting a vibrant online community with diverse new features

Dcard has always been committed to enabling more interactions among users and creating strong bonds within the online community. The team is now adding small games to the forum allowing users to play against each other. Next, Dcard plans to leverage Vertex AI (https://cloud.google.com/vertex-ai/) to develop features powered by artificial intelligence (AI) that can further enhance user engagement. It will also integrate the large language model of Google with its data pipeline centered around BigQuery to provide automatically summarized product reviews in its ecommerce service, reinforce its recommendation system, and analyze its query records in BigQuery to optimize its data marts.

Taco says, "Our user base is rather young and vibrant, so we're constantly creating new features to cater to their needs. With its high-performance infrastructure and diverse cloud tools, Google Cloud has supported our rapid growth over the past years. We're confident that we'll be able to continue expanding and strengthening our online community using Google Cloud."

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